



Function Flavors Overview

Brief Overview

This note covers **JavaScript functions** and was created from an [uploaded document](#). It explains vanilla, parameterized, and money-based function styles, uses `Math.floor()`, and compares usage scenarios.

Key Points

- Vanilla functions: no parameters, fixed behavior.
 - Parameterized functions: accept arguments, flexible outputs.
 - Money-based functions: calculate quantities from a given budget.
 - `Math.floor()`: rounds numbers down to whole items.
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Function Flavors Overview



Vanilla – function without parameters

Function: a named block of code that runs when the name is invoked.

- Defined with the function keyword.
- No values are passed between the parentheses ().
- Example call: `getMilk()` – runs the same set of commands each time.



Chocolate – function with **parameters**

Parameter: a variable listed inside the parentheses of a function definition; it receives a value when the function is called.

- Allows the function to work with different data each call.
- Syntax: `function functionName(parameter) { ... }`

Example: `getMilk(bottles)`

```
function getMilk(bottles) {  
  console.log("Buy " + bottles + " bottles of milk");  
  var cost = bottles * 1.5;           // $1.5 per bottle  
}
```

```
    console.log("Cost: $" + cost);  
}
```

- Calling `getMilk(2)` binds the value 2 to the parameter bottles.
- Calculations:
 - Number of bottles: 2
 - Cost: $2 \times 1.5 = 3$
 - Output:

```
Buy 2 bottles of milk  
Cost: $3
```

Extending the function – input as money

`Math.floor()`: JavaScript method that rounds a number **down** to the nearest integer.

```
function getMilk(money) {  
    var numberOfBottles = Math.floor(money / 1.5);  
    console.log("Buy " + numberOfBottles + " bottles of milk");  
}
```

- money receives the amount supplied when calling the function.
- numberOfBottles is computed by dividing money by the unit price \$1.5 and rounding down.
- Calling `getMilk(5)` yields:
 - Calculation: $5 \div 1.5 = 3.\overline{3}$
 - After `Math.floor`, 3
 - Output:

```
Buy 3 bottles of milk
```

Comparison of Function Flavors

Flavor	Parameters	Typical Use	Example Call
Vanilla	None	Fixed behavior	<code>getMilk()</code>

Chocolate	One or more values (e.g., bottles)	Variable behavior based on input	getMilk(2)
Money-based	Input represents money amount	Compute quantity from budget	getMilk(5)

Key Takeaways

- **Parameters** make functions reusable with different data.
 - Use arithmetic inside functions to compute results (e.g., cost, quantity).
 - Apply **Math.floor()** when you need to discard fractional parts, such as counting whole items.
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